

AI Assisted Coding

ASSIGNMENT.6.1

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[Batch-14]

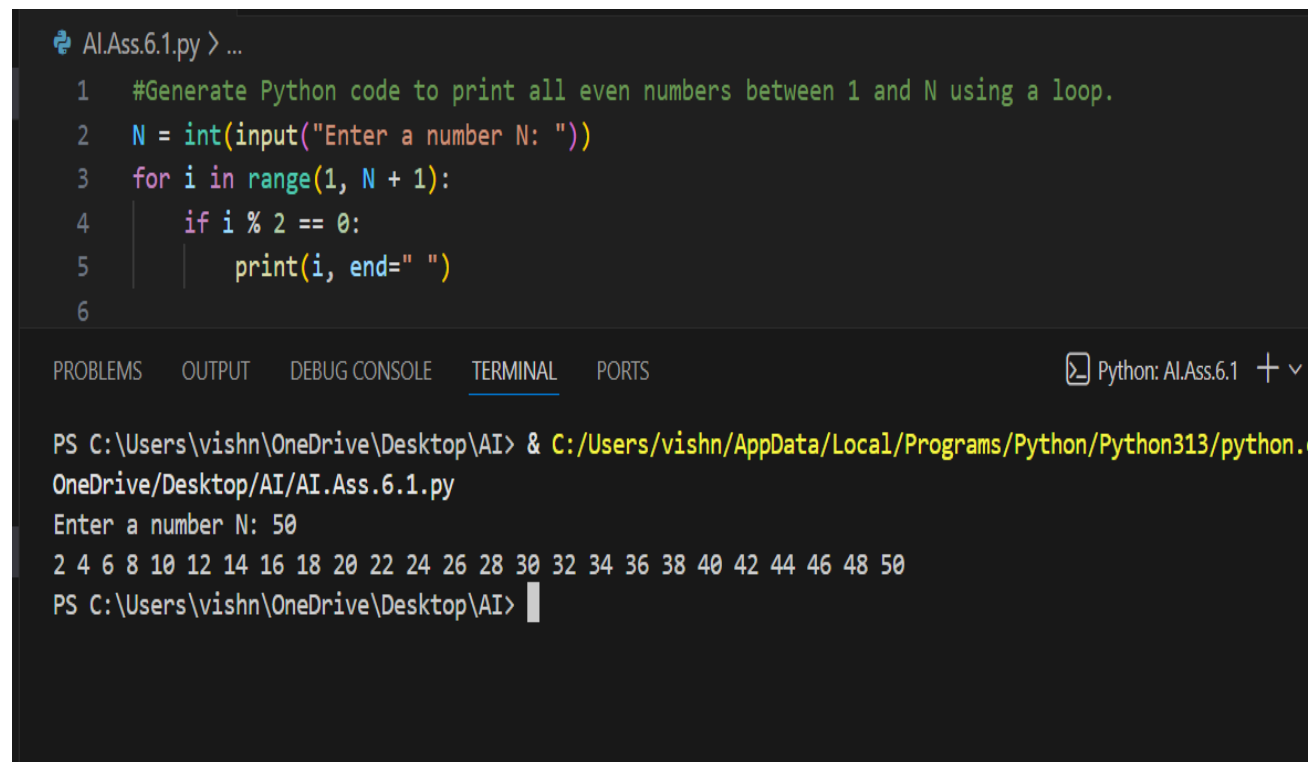
*** AI-Based Code Completion: Working with suggestions for classes, loops, conditionals**

Task 1:

Prompt:

Generate Python code to print all even numbers between 1 and N using a loop.

Code&Output:



```
AI.Ass.6.1.py > ...
1  #Generate Python code to print all even numbers between 1 and N using a loop.
2  N = int(input("Enter a number N: "))
3  for i in range(1, N + 1):
4      if i % 2 == 0:
5          print(i, end=" ")
6

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: AI.Ass.6.1 + v
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.
OneDrive/Desktop/AI/AI.Ass.6.1.py
Enter a number N: 50
2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50
PS C:\Users\vishn\OneDrive\Desktop\AI> 
```

Task 2:

Prompt:

Generate Python code to count how many numbers in a list are even and odd.

Code&Output:

```
AI.Ass.6.1.py > ...
7 #Generate Python code to count how many numbers in a list are even and odd.
8 numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
9 even_count = 0
10 odd_count = 0
11 for num in numbers:
12     if num % 2 == 0:
13         even_count += 1
14     else:
15         odd_count += 1
16 print(f"Even numbers: {even_count}, Odd numbers: {odd_count}")
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: AI.Ass.6.1 + v

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.
OneDrive/Desktop/AI/AI.Ass.6.1.py
Even numbers: 5, Odd numbers: 5
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Task 3:

Prompt:

Generate a Python class User that validates age and email using conditional statements.

Code:

```
AI.Ass.6.1.py > ...
18 #Generate a Python class User that validates age and email using conditional statements with user input
19 class User:
20     def __init__(self, age, email):
21         self.age = age
22         self.email = email
23
24     def validate_age(self):
25         if self.age < 18:
26             return False
27         elif self.age > 150:
28             return False
29         else:
30             return True
31
32     def validate_email(self):
33         if "@" not in self.email:
34             return False
35         elif "." not in self.email:
36             return False
37         else:
38             return True
39
40 # User input
41 age = int(input("Enter your age: "))
42 email = input("Enter your email: ")
43 user = User(age, email)
44 if user.validate_age():
45     print("Valid age")
46 else:
47     print("Invalid age")
48 if user.validate_email():
49     print("Valid email")
50 else:
51     print("Invalid email")
```

Output :

```
Python: AI.Ass.6.1 + v
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe AI.Ass.6.1.py
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe AI.Ass.6.1.py
OneDrive/Desktop/AI/AI.Ass.6.1.py
Enter your age: 22
Enter your email: vishnu@com.
Valid age
Valid email
PS C:\Users\vishn\OneDrive\Desktop\AI> 
```

Task 4:

Prompt:

#Generate a Python class Student with attributes (name, roll number,marks) and methods to calculate total and average marks

Code&Output:

```
AI.Ass.6.1.py > ...
51 #Generate a Python class Student with attributes (name, roll number,marks) and methods to calcul
52 class Student:
53     def __init__(self, name, roll_number, marks):
54         self.name = name
55         self.roll_number = roll_number
56         self.marks = marks # marks should be a list of integers
57
58     def calculate_total(self):
59         return sum(self.marks)
60
61     def calculate_average(self):
62         if len(self.marks) == 0:
63             return 0
64         return self.calculate_total() / len(self.marks)
65 # Example usage
66 name = input("Enter student name: ")
67 roll_number = input("Enter roll number: ")
68 marks = list(map(int, input("Enter marks separated by space: ").split()))
69 student = Student(name, roll_number, marks)
70 total_marks = student.calculate_total()
71 average_marks = student.calculate_average()
72 print(f"Total Marks: {total_marks}")
73 print(f"Average Marks: {average_marks}")
74
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python: AI.Ass.6.1 + v []

```
PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c
OneDrive/Desktop/AI/AI.Ass.6.1.py
Enter student name: vishnu
Enter roll number: 10A0
Enter marks separated by space: 97 99 89
Total Marks: 285
Average Marks: 95.0
PS C:\Users\vishn\OneDrive\Desktop\AI>
```

Task 5:

Prompt:

#Generate a Python program for a simple bank account system using class, loops, and conditional statements.

Code:

```
Al.Ass.6.1.py > BankAccount > get_balance

76 #Generate a Python program for a simple bank account system using class, loops, and conditional statements.
77 class BankAccount:
78     def __init__(self, account_number, initial_balance=0):
79         self.account_number = account_number
80         self.balance = initial_balance
81     def deposit(self, amount):
82         if amount > 0:
83             self.balance += amount
84             print(f"Deposited ${amount}. New balance: ${self.balance}")
85         else:
86             print("Invalid deposit amount.")
87     def withdraw(self, amount):
88         if 0 < amount <= self.balance:
89             self.balance -= amount
90             print(f"Withdrew ${amount}. New balance: ${self.balance}")
91         else:
92             print("Invalid withdrawal amount or insufficient funds.")
93     def get_balance(self):
94         return self.balance
95 account = BankAccount("123456789", 1000)
96 account.deposit(500)
97 account.withdraw(200)
98 print(f"Final balance: ${account.get_balance()}")
99 while True:
100     action = input("Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: ").lower()
101     if action == 'd':
102         amount = float(input("Enter amount to deposit: "))
103         account.deposit(amount)
104     elif action == 'w':
105         amount = float(input("Enter amount to withdraw: "))
106         account.withdraw(amount)
107     elif action == 'b':
108         print(f"Current balance: ${account.get_balance()}")
109     elif action == 'q':
110         print("Exiting the banking system.")
111         break
112     else:
113         print("Invalid option. Please try again.")
114
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\vishn\OneDrive\Desktop\AI> & C:/Users/vishn/AppData/Local/Programs/Python/Python313/python.exe c:/Users/v
Deposited $500. New balance: $1500
Withdrew $200. New balance: $1300
Final balance: $1300
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: d
Enter amount to deposit: 23456
Deposited $23456.0. New balance: $24756.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Final balance: $1300
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: d
Enter amount to deposit: 23456
Deposited $23456.0. New balance: $24756.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: d
Enter amount to deposit: 23456
Deposited $23456.0. New balance: $24756.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Enter amount to deposit: 23456
Deposited $23456.0. New balance: $24756.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Deposited $23456.0. New balance: $24756.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Enter amount to withdraw: 7656
Withdrew $7656.0. New balance: $17100.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: b
Current balance: $17100.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: 
```